



Heart Disease

Donated on 6/30/1988

4 databases: Cleveland, Hungary, Switzerland, and the VA Long Beach

Dataset Characteristics

Multivariate

Subject Area

Health and Medicine

Associated Tasks

Classification

Feature Type

Categorical, Integer, Real

Instances

303

Features

13



DOWNLOAD (125.9 KB)



IMPORT IN PYTHON

CITE

64 citations

791940 views

Keywords

health

Creators



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DOI

Dataset Information

Additional Information

This database contains 76 attributes, but all published experiments refer to using a subset of 14 of them. In particular, the Cleveland database is the only one that has been used by ML researchers to date. The "goal" field refers to the presence of heart disease in the patient. It is integer valued from 0 (no presenc...

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Has Missing Values?

Yes

Introductory Paper



[International application of a new probability algorithm for the diagnosis of coronary artery disease.](#)

By R. Detrano, A. Jánosi, W. Steinbrunn, M. Pfisterer, J. Schmid, S. Sandhu, K. Guppy, S. Lee, V. Froelicher. 1989
Published in American Journal of Cardiology

10.24432/C52P4X

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This allows for the sharing and adaptation of the datasets for any purpose, provided that the appropriate credit is given.

Variables Table



Variable Name	Role	Type	Demographic	Description	Units	Missing Values
age	Feature	Integer	Age		years	no
sex	Feature	Categorical	Sex			no
cp	Feature	Categorical				no
trestbps	Feature	Integer		resting blood pressure (on admission to the hospital)	mm Hg	no
chol	Feature	Integer		serum cholestoral	mg/dl	no
fbs	Feature	Categorical		fasting blood sugar > 120 mg/dl		no
restecg	Feature	Categorical				no

Variable Name	Role	Type	Demographic	Description	Units	Missing Values
thalach	Feature	Integer		maximum heart rate achieved		no
exang	Feature	Categorical		exercise induced angina		no
oldpeak	Feature	Integer		ST depression induced by exercise relative to rest		no
slope	Feature	Categorical				no
ca	Feature	Integer		number of major vessels (0-3) colored by flourosopy		yes
thal	Feature	Categorical				yes
num	Target	Integer		diagnosis of heart disease		no

Rows per page 25 0 to 14 of 14 < >

Additional Variable Information



Only 14 attributes used:

1. #3 (age)
2. #4 (sex)
3. #9 (cp)
4. #10 (trestbps)
5. #12 (chol)
6. #16 (fbs)
7. #19 (restecg)
8. #32 (thalach)
9. #38 (exang)
10. #40 (oldpeak)
11. #41 (slope)
12. #44 (ca)
13. #51 (thal)
14. #58 (num) (the predicted attribute)

Complete attribute documentation:

- 1 id: patient identification number
- 2 ccf: social security number (I replaced this with a dummy value of 0)
- 3 age: age in years
- 4 sex: sex (1 = male; 0 = female)
- 5 painloc: chest pain location (1 = substernal; 0 = otherwise)
- 6 painexer (1 = provoked by exertion; 0 = otherwise)
- 7 relrest (1 = relieved after rest; 0 = otherwise)
- 8 pncaden (sum of 5, 6, and 7)
- 9 cp: chest pain type
 - Value 1: typical angina
 - Value 2: atypical angina
 - Value 3: non-anginal pain

-- Value 4: asymptomatic

10 trestbps: resting blood pressure (in mm Hg on admission to the hospital)

11 htn

12 chol: serum cholestoral in mg/dl

13 smoke: I believe this is 1 = yes; 0 = no (is or is not a smoker)

14 cigs (cigarettes per day)

15 years (number of years as a smoker)

16 fbs: (fasting blood sugar > 120 mg/dl) (1 = true; 0 = false)

17 dm (1 = history of diabetes; 0 = no such history)

18 famhist: family history of coronary artery disease (1 = yes; 0 = no)

19 restecg: resting electrocardiographic results

-- Value 0: normal

-- Value 1: having ST-T wave abnormality (T wave inversions and/or ST elevation or depression of > 0.05 mV)

-- Value 2: showing probable or definite left ventricular hypertrophy by Estes' criteria

20 ekgmo (month of exercise ECG reading)

21 ekgday(day of exercise ECG reading)

22 ekgyr (year of exercise ECG reading)

23 dig (digitalis used during exercise ECG: 1 = yes; 0 = no)

24 prop (Beta blocker used during exercise ECG: 1 = yes; 0 = no)

25 nitr (nitrates used during exercise ECG: 1 = yes; 0 = no)

26 pro (calcium channel blocker used during exercise ECG: 1 = yes; 0 = no)

27 diuretic (diuretic used during exercise ECG: 1 = yes; 0 = no)

28 proto: exercise protocol

1 = Bruce

2 = Kottus

3 = McHenry

4 = fast Balke

5 = Balke

- 6 = Noughton
- 7 = bike 150 kpa min/min (Not sure if "kpa min/min" is what was written!)
- 8 = bike 125 kpa min/min
- 9 = bike 100 kpa min/min
- 10 = bike 75 kpa min/min
- 11 = bike 50 kpa min/min
- 12 = arm ergometer
- 29 thaldur: duration of exercise test in minutes
- 30 thaltime: time when ST measure depression was noted
- 31 met: mets achieved
- 32 thalach: maximum heart rate achieved
- 33 thalrest: resting heart rate
- 34 tpeakbps: peak exercise blood pressure (first of 2 parts)
- 35 tpeakbpd: peak exercise blood pressure (second of 2 parts)
- 36 dummy
- 37 trestbpd: resting blood pressure
- 38 exang: exercise induced angina (1 = yes; 0 = no)
- 39 xhypo: (1 = yes; 0 = no)
- 40 oldpeak = ST depression induced by exercise relative to rest
- 41 slope: the slope of the peak exercise ST segment
 - Value 1: upsloping
 - Value 2: flat
 - Value 3: downsloping
- 42 rldv5: height at rest
- 43 rldv5e: height at peak exercise
- 44 ca: number of major vessels (0-3) colored by flourosopy
- 45 restckm: irrelevant
- 46 exerckm: irrelevant
- 47 restef: rest raidonuclid (sp?) ejection fraction

48 restwm: rest wall (sp?) motion abnormality
0 = none
1 = mild or moderate
2 = moderate or severe
3 = akinesis or dyskmem (sp?)
49 exeref: exercise radinalid (sp?) ejection fraction
50 exerwm: exercise wall (sp?) motion
51 thal: 3 = normal; 6 = fixed defect; 7 = reversable defect
52 thalsev: not used
53 thalpul: not used
54 earlobe: not used
55 cmo: month of cardiac cath (sp?) (perhaps "call")
56 cday: day of cardiac cath (sp?)
57 cyr: year of cardiac cath (sp?)
58 num: diagnosis of heart disease (angiographic disease status)
-- Value 0: < 50% diameter narrowing
-- Value 1: > 50% diameter narrowing
(in any major vessel: attributes 59 through 68 are vessels)
59 lmt
60 ladprox
61 laddist
62 diag
63 cxmain
64 ramus
65 om1
66 om2
67 rcaprox
68 rcadist
69 lvx1: not used

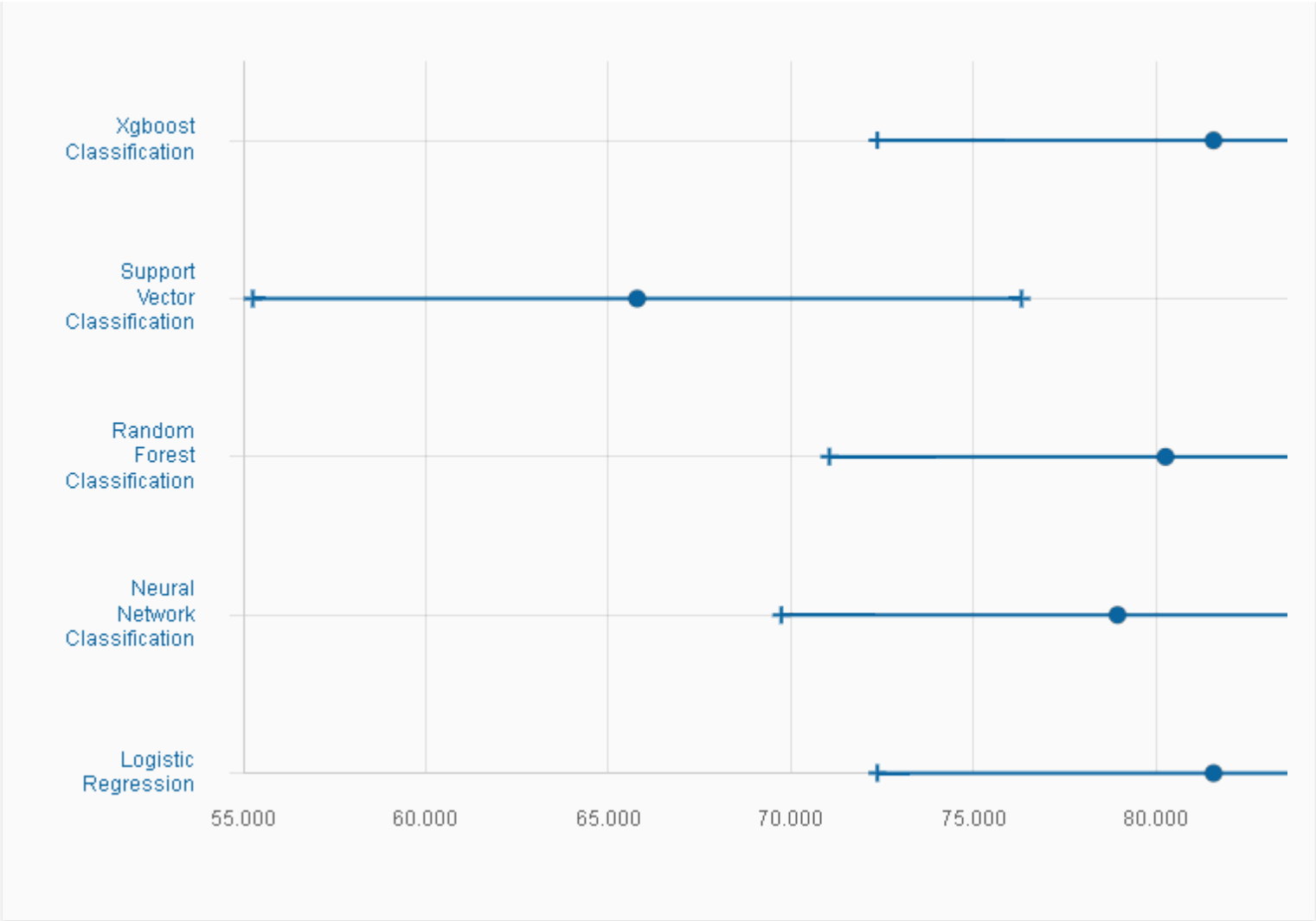
70 lvx2: not used
71 lvx3: not used
72 lvx4: not used
73 lvf: not used
74 cathef: not used
75 junk: not used
76 name: last name of patient (I replaced this with the dummy string "name")

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Baseline Model Performance ^

Accuracy

Precision



Dataset Files



File	Size
new.data	380.6 KB

File	Size
hungarian.data	60.7 KB
cleveland.data	59.2 KB
long-beach-va.data	39 KB
switzerland.data	24.1 KB

Rows per page 5 ▾ 0 to 5 of 22

Papers Citing this Dataset

 SORT BY YEAR, DESC

[A Submodularity-based Agglomerative Clustering Algorithm for the Privacy Funnel](#)

By Ni Ding, Parastoo Sadeghi. 2019

Published in ArXiv.

[Leveraging Simple Model Predictions for Enhancing its Performance](#)

By Amit Dhurandhar, Karthikeyan Shanmugam, Ronny Luss. 2019

Published in ArXiv.

[PPCD: Privacy-preserving clinical decision with cloud support](#)

By Hui Ma, Xuyang Guo, Yuan Ping, Baocang Wang, Yuehua Yang, Zhili Zhang, Jingxian Zhou. 2019

Published in PloS one.

[ActShare: Sensitive Data Sharing with Reliable Leaker Identification](#)

By Zhiyi Zhang, Yu Guan, Xinyu Ma, Lixia Zhang. 2019
Published in ArXiv.

[A review of domain adaptation without target labels](#)

By Wouter Kouw, Marco Loog. 2019
Published in

Rows per page

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0 to 5 of 64

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Reviews

5.0



(4 ratings)

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... Thank You ... Mucho Gusto e Gracias

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